UNIT 5

DISTANCE, TIME AND TEMPERATURE

DISTANCE

Table of relationship among units of length.

10 millimetres (mm)	=	1 centimetre (cm)
100 centimetres (cm)	=	1 metre (m)
1000 metres (m)	=	1 kilometre (km)

Conversion of kilometres to metres and metres to kilometres

To convert kilometres to metres, we multiply the number of kilometres by 1000.

Examples: (1) Convert 5 km to metres

(2) Convert 8 km 150 m to metres

Solution:

Distance = 5 km

= (5 x 1000) m

= 5000 m

Solution:

Distance 8 km 150 m

= (8 x 1000) m + 150 m

= 8000 m + 150 m

= 8150 m

Examples:

(1) Convert 12000 m to km

Solution:

12000 m = (12000 ÷ 1000) km

$$= \frac{12\cancel{000}}{1\cancel{000}} \text{ km} = \frac{12}{1} \text{ km} = 12 \text{ km}$$

(2) Convert 18450 m to km and m

Solution:

18450 m = 18000 m + 450 m

$$= \frac{\cancel{18000}}{\cancel{1000}} \text{ km } + 450 \text{ m}$$

= 18 km 450 m

$$\begin{array}{r|l} & \textbf{18 km} \\ \hline 1000 & 18450 \\ & -\ 1000 \\ \hline & 8450 \\ & -\ 8000 \\ \hline & \textbf{450 m} \end{array}$$

Conversion of metres to centimetres and centimetres to metres

To convert metres to centimetres, we multiply the number of centimeters by 100.

Examples: (1) Convert 11 m to cm

(2) Convert 15 m 30 cm to cm

Solution:

11 m = (11 x 100) cm
= 1100 cm

Solution:

15 m 30 cm
= (15 x 100) cm + 30 cm
= 1500 cm + 30 cm
= 1530 cm

Examples:

(1) Convert 1400 cm to m

Solution:

1400 cm = (1400 ÷ 100) m

$$= \overset{14}{\cancel{1400}} \times \frac{1}{\underset{1}{\cancel{100}}} \text{ m} = 14 \text{ m}$$

(1) Convert 2436 cm to m and cm

Solution:

2436 cm = 2400 cm + 36 cm

= (2400 ÷ 100) m + 36 cm

$$= \left(\overset{24}{\cancel{2400}} \times \frac{1}{\underset{1}{\cancel{100}}}\right) \text{ m} + 36 \text{ cm} = 24 \text{ m } 36 \text{ cm}$$

Conversion of centimetres to millimetres and millimetres to centimetres

To convert centimetres to millimetres, we multiply the number of centimetres by 10.

Examples: (1) Convert 16 cm to mm

(2) Convert 25 cm 4 mm to mm

Solution:

16 cm = (16 x 10) mm

= 160mm

Solution:

25 cm 4 mm

= 25 cm + 4 mm

= (25 x 10) mm + 4 mm

= 250 mm + 4 mm = 254 mm

Examples:

(3) Convert 350 mm to cm

Solution:

350 mm = (350 ÷ 10) cm

$$\left(\overset{35}{\cancel{350}} \times \frac{1}{\underset{1}{\cancel{10}}}\right) \text{ cm} = 35 \text{ cm}$$

(4) Convert 2436 cm to m and cm

Solution:

145 mm

= (145 ÷ 10) cm

= 14.5 cm

= 14 cm 5 mm

EXERCISE 5.1

Q1. Change into metres.

1) 550 cm = 　　 **2)** 950 cm =

3) 195 cm = 　　 **4)** 330 cm =

5) 830 cm = 　　 **6)** 105 cm =

Q2. Change into centimetres.

1) 1 m 25 cm =

2) 2 m 30 cm =

3) 4 m 10 cm =

4) 0 m 59 cm =

5) 9 m 99 cm =

6) 3 m 68 cm =

Q3. Change the following.

1) 1 km = ______ cm

2) 5 m = ______ cm

3) 7 m = ☐ cm

4) 7 m 42 cm = ☐ cm

5) 2 km 525 m = ☐ cm

6) ☐ 500 m = 3500 m

Q4. Convert km into metres.

1) 8 km = ☐ m

2) 31 km = ☐ m

3) 5 km = ☐ m

4) 103 km = ☐ m

5) 12 km = ☐ m

6) 306 km = ☐ m

7) 29 km = ☐ m

8) 81 km = ☐ m

Q5. Express in metres.

1) 2 km = ☐ m

2) 4 km = ☐ m

3) 5 km = ☐ m

4) 8 km = ☐ m

Q6. Express in kilometres.

1) 3000 m = ☐ km

2) 6000 m = ☐ km

3) 7000 m = ☐ km

4) 9000 m = ☐ km

Q7. Express in metres.

1) 1 km 145 m = ☐ m

2) 1 km 145 m = ☐ m

3) 1 km 298 m = ☐ m

4) 2 km 78 m = ☐ m

5) 2 km 580 m = ☐ m

6) 1 km 6 m = ☐ m

7) 3 km 670 m = ☐ m

ADDITION

Example.

Add 3m 35cm distance to 8m 75cm.

Solution:

	m	cm
	1 3	1 35
+	8	75
	12	10

Therefore, addition of distances: 3 m 35 cm + 8 m 75 cm = 12 m 10 cm

EXERCISE 5.2

Q1. Add the following.

	m	cm
	12	40
+	18	10

	m	cm
	15	20
+	11	15

	m	cm
	10	54
+	49	69

Q2. Add the following.

	m	cm
	31	40
+	20	50

	m	cm
	29	22
+	10	11

	m	cm
	2	81
+	8	11

	m	cm
	30	25
+	15	45

	m	cm
	96	37
+	00	43

	m	cm
	10	20
+	58	63

	m	cm
	51	15
+	45	30

	m	cm
	62	20
+	26	02

	m	cm
	11	90
+	88	00

Q3. Add the following.

	km	m
	5	3
+	8	4

	km	m
	25	105
+	22	203

	km	m
	1500	500
+	2000	700

	km	m
	33	38
+	42	47

	km	m
	92	55
+	50	45

	km	m
	49	35
+	40	95

	m	cm
	300	200
+	450	550

	m	cm
	500	200
+	450	300

	m	cm
	80	27
+	21	56

Q4. Add the following.

1) 12 km + 33 km = ______ km

2) 231 cm + 455 cm = ______ cm

3) 4300 m + 2323 m = ______ m

4) 67 mm + 88 mm = ______ mm

5) 6.25 km + 5.03 km = ______ km

6) 56.45 m + 213.1 m = ______ m

7) 7.88 cm + 9.67 cm = ______ cm

8) 322 mm + 69 mm = ______ mm

9) 71.34 m + 35.50 m = ______ m

10) 100 km + 100 km = ______ km

SUBTRACTION

Example.

Subtract 4m 80cm from 11m 15cm distances.

Solution:

	m	cm
	10	1
	1~~1~~	15
-	4	80
	6	35

EXERCISE 5.3

Q1. Subtract the following.

	m	cm
	14	39
-	3	14

	m	cm
	23	75
-	12	62

	m	cm
	8	15
-	3	20

Q2. Subtract the following.

	km	m
	499	700
-	300	250

	km	m
	849	300
-	441	200

	km	m
	240	100
-	135	150

	m	cm
	50	80
-	30	75

	m	cm
	42	50
-	35	37

	m	cm
	250	90
-	242	45

Q3. Subtract the following.

1) 48 m 14 cm - 41 m 7 cm =

2) 20 m 98 cm - 15 m 62 cm =

3) 4 m 8 cm - 3 m 3 cm =

4) 15 m 11 cm - 10 m 8 cm =

5) 26 m 85 cm - 23 m 46 cm =

6) 33 m 58 cm - 9 m 13 cm =

7) 27 m 35 cm - 20 m 34 cm =

8) 22 m 40 cm - 11 m 8 cm =

WORD PROBLEMS

1 The length of a ribbon is 6m 80cm. How much ribbon is left if 2m 88cm has been cut off?

2 Jameel covered a distance of 589m from his house to Jamia Masjid and then 868m from Jamia Masjid to school. Find the total distance covered by him.

3 A car is 1m 62cm wide. A garage is 2m 41cm wide. How much space is left when the car is in the garage?

TIME

Table of relationship among units of length.

24 hours	=	1 day
7 days	=	1 week
4 weeks	=	1 month
1 month	=	30 days
12 months	=	1 year
365 days	=	1 year

Examples: (1) Convert 600 minutes to hours

(2) Convert 400 minutes to hours and minutes

Solution:

600 minutes

600 ÷ 60 hours

= 10 hours

Solution:

400 minutes

= (400 ÷ 60) hours

= 6 hours 40 minutes

$$\begin{array}{r|r} & 6 \\ \hline 60 & 400 \\ & -360 \\ \hline & 40 \end{array}$$

Examples: Convert 660 seconds into minutes

Solution:

Given 660 seconds
= (660 ÷ 60) minutes
= 11 minutes

A. Convert years to months and months to years

To convert years into months, we multiply the number of years by 12.

Example:

(a) Convert 11 years to months
(b) Convert 5 years 10 months to months

Solution:

(a) 11 years = (11 × 12) months = 132 months

(b) 5 years 10 months = 5 years + 10 months
= (5 × 12) months + 10 months
= 60 months + 10 months
= 70 months

B. Convert months to years

To convert months to years, we divide the number of months by 12.

Example:

(a) Convert 48 months to years
(b) Convert 81 months to years and months

Solution:

(a) 48 months = (48 ÷ 12) years = 4 years

(b) 81 months = (81 ÷ 12) years = (72 ÷ 12) years + 9 months
= 6 years 9 months

C. Convert months to days

To convert months to days, we multiply the number of months by 30.

Example:

(a) Convert 18 months to days

(b) Convert 13 months 25 days to days

Solution:

(a) 18 months = 18×30 days = 540 days

(b) 13 months 25 days = 13 months + 25 days

= (13×30) days + 25 days

= 390 days + 25 days = 415 days

D. Convert days to months

To convert days to months, we divide the number of days by 30.

Example:

(a) Convert 150 days to months

(b) Convert 244 days to months and days

Solution:

(a) 150 days = $(150 \div 30)$ months = 5 months

(b) 244 days = $(244 \div 30)$ months

= $(240 \div 30)$ months + 4 days

= 8 months 4 days

E. Convert weeks to days

To convert weeks to days, we multiply the number of weeks by 7.

Example:

(a) Convert 4 weeks to days

(b) Convert 14 weeks 2 days to days

Solution:

(a) 4 weeks = 4×7 days = 28 days

(b) 14 weeks 2 days = 14 weeks + 2 days
= (14 × 7) days + 2 days
= 98 days + 2 days = 100 days

F. Convert days to weeks

To convert days into weeks, we divide the number of days by 7.

Example:

(a) Convert 98 days to weeks

(b) Convert 125 days to weeks and days

Solution:

(a) 98 days = (98 ÷ 7) weeks = 14 weeks

$$\begin{array}{r} 17 \\ 7\overline{)125} \\ -7 \\ \hline 55 \\ -49 \\ \hline 6 \end{array}$$

(b) 125 days = (125 ÷ 7) weeks
(Divide 125 ÷ 7, we get 17 weeks and remainder 6 days)
= 17 weeks 6 days

EXERCISE 5.4

Q1. Convert from days to hours.

1) 7 days = ☐ hour

2) 1 days = ☐ hour

3) 11 days = ☐ hour

4) 3 days = ☐ hour

5) 5 days = ☐ hour

6) 9 days = ☐ hour

7) 13 days = ☐ hour

8) 15 days = ☐ hour

Q2. Convert from hours to days.

1) 96 hours = ☐ days

2) 96 hours = ☐ days

3) 192 hours = ☐ days

4) 384 hours = ☐ days

5) 240 hours = ☐ days

6) 48 hours = ☐ days

7) 336 hours = ☐ days

8) 288 hours = ☐ days

Q3. Convert the following.

1) 20 minutes = ☐ seconds

2) 18 days = ☐ hours

3) 600 minutes = ☐ hours

4) 5 days = ☐ minutes

5) 2352 hours = ☐ weeks

6) 17 hours = ☐ minutes

7) 17 weeks = ☐ days

8) 360 hours = ☐ days

Q4. Answer the following question.

1) 1 week = ☐ days

2) 2 minutes = ☐ seconds

3) 1 day = ☐ hours

4) 1 fortnight = ☐ weeks

5) 1 year = ☐ months

6) 3 minutes = ☐ seconds

7) 4 hours = ☐ minutes

8) 3 years = ☐ months

ADDITION

Example.

Add 3 hours 55 minutes and 5 hours 40 minutes.

Solution:

	h	m
	①3	55
+	5	40
	9	35

55 minutes + 40 minutes = 95 minutes
= 1 hour 35 minutes
Write 35 minutes under the minutes column and carry 1 to the hour's column.
Thus, **9** hours **35** minutes.

EXERCISE 5.5

Q1. Add the following.

Example:

	12 hr.	45 min	29 sec
+	60 hr.	19 min	47 sec
	19 hr	24 min	16 sec

	14 hr.	03 min	04 sec
+	04 hr.	19 min	09 sec

	07 hr.	56 min	14 sec
+	01 hr.	51 min	51 sec

	22 hr.	31 min	36 sec
+	20 hr.	35 min	41 sec

	01 hr.	55 min	43 sec
+	18 hr.	32 min	39 sec

Q2. Add the following.

	1 hour	32 minutes
+	2 hours	17 minutes

	3 hours	22 minutes
+	6 hours	11 minutes

	2 hours	20 minutes
+	5 hours	18 minutes

	4 hours	2 minutes
+	13 hours	20 minutes

Q3. Answer the following questions.

1) 45 min + 3 h 5 min = ☐ h ☐ min

2) 50 min + 1 h 50 min = ☐ h ☐ min

3) 45 min + 3 h 40 min = ☐ h ☐ min

4) 2 h 20 min + 50 min = ☐ h ☐ min

5) 2 h 40 min + 2 h 5 min = ☐ h ☐ min

SUBTRACTION

Example.

Subtract 58 minutes 15 second from 72 minutes and 30 seconds.

Solution:

m	s
⑥ ① 72	⑩ 15
− 58	30
13	45

15 seconds - 30 seconds
We borrow 1 minute = 60 seconds
So, 60 + 15 = 75 seconds
75 - 30 = 45 seconds
Thus, **13** minutes **45** seconds

EXERCISE 5.6

Q1. Subtract the following.

	22 hours 56 minutes
−	9 hours 11 minutes

	15 hours 26 minutes
−	7 hours 11 minute

	16 hours 42 minutes
−	5 hours 18 minutes

	3 hours 39 minutes
−	1 hour 10 minutes

Q2. Add or subtract each pair of times.

	1 hour	10 minutes	2 seconds
+	22 hours	21 minutes	41 seconds

	16 hours	39 minutes	34 seconds
–	12 hours	9 minutes	16 seconds

	17 hours	34 minutes	28 seconds
–	14 hours	34 minutes	20 seconds

	7 hours	28 minutes	18 seconds
+	3 hours	27 minutes	12 seconds

	15 hours	12 minutes	1 second
+	6 hours	36 minutes	49 seconds

	13 hours	54 minutes	10 seconds
–	10 hours	30 minutes	10 seconds

	9 hours	42 minutes	37 seconds
–	8 hours	7 minutes	13 seconds

WORD PROBLEMS

1 Umair plays for 1 hour 15 minutes, and Ali plays for 55 minutes. Who plays for a shorter period, and by how much?

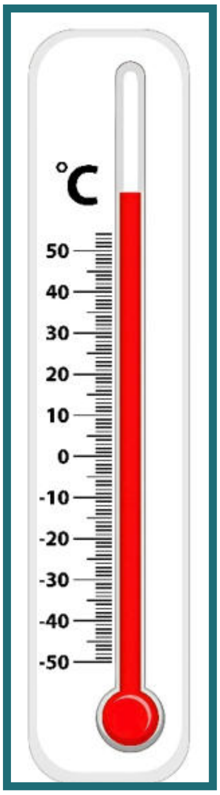
2 Fatima takes 1 hour 20 minutes to complete her Mathematics homework and 50 minutes to complete her English homework. How much total time does she take to complete the homework for both subjects?

3 School closes for summer vacations for a period of 2 months and 5 days, while the other holidays last for 27 days. How many days will the school remain closed?

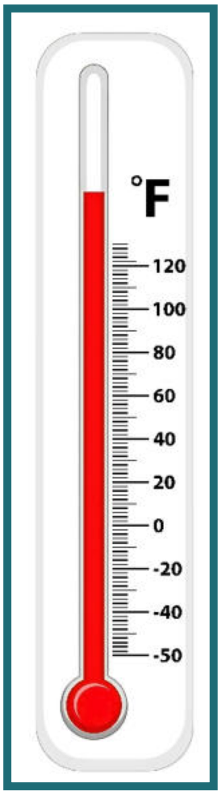
TEMPERATURE

The instrument used to measure temperature is called thermometer.
There are two types of scales used in thermometers.

1. Celsius scale



2. Fahrenheit scale



Conversion of Fahrenheit and Celsius

Temperature Helpers

$$°F = °C \times \frac{9}{5} + 32$$

$$°C = (°F - 32) \times \frac{5}{9}$$

(i) Converting Temperature From Fahrenheit Scale To Celsius Scale.

Example: Convert 59°F to the Celsius scale.
Solution:

Step 1: Subtraction of 32 from 59 gives

$$59 - 32 = 27$$

Step 2: Multiplication of 27 by $\frac{5}{9}$ gives

$$\overset{3}{\cancel{27}} \times \frac{5}{\cancel{9}} = 3 \times 5 = 15$$

Thus, 59°F = 15°C.

(ii) Converting Temperature From Celsius Scale To Fahrenheit Scale.

Example: Convert 35°C to the Fahrenheit scale.
Solution:

Step 1: Multiplication of given temperature by $\frac{9}{5}$ gives

$$\overset{3}{\cancel{35}} \times \frac{9}{\cancel{5}} = 7 \times 9 = 63$$

Step 2: Addition of 32 and 63 gives

$$63+32=95$$

Thus, 35°C = 95°F.

EXERCISE 5.7

Q1. Convert Fahrenheit to Celsius.

1. 39°F

2. 83°F

3. 65°F

4. 62°F

5. 94°F

6. 111°F

7. 34°F

8. 38°F

Q2. Convert Celsius to Fahrenheit.

1. 44°C

2. 38°C

3. 9°C

4. 51°C

5. 45°C

6. 12°C

7. 23°C

8. 21°C

Q3. Solve the following.

1. −44°C °F

2. 75°C °F

3. −74°C °F

4. 39°C °F

5. 90°C °F

6. 56°C °F

7. 53°C °F

8. −46°C °F

8. −11°C °F

Q4. Convert the following.

1. 23°F °C

2. −10°F °C

3. 87°F °C

4. 110°F ______ °C

5. 34°C ______ °F

6. 31°C ______ °F

WORD PROBLEMS

1 The maximum temperature on a day in Hyderabad was 35°C. What was the temperature in degrees Fahrenheit?

2 During summer, on one day, the temperature in Jacobabad was 113°F, whereas the temperature in Hyderabad was 40°C. Which city had a higher temperature, and by how much?

3 Ali was suffering from fever. His temperature at noon was 104°F. Convert the said temperature to degrees Celsius.

4 In summer, on one day, the maximum temperature in Sukkur was 104°F, whereas in Karachi, it was 35°C. What was the difference in the temperature of both cities in degrees Fahrenheit?